In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
import warnings
warnings.filterwarnings('ignore')
```

```
C:\Users\Prashant\anaconda3\lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
  from pandas.core import (
```

In [2]:
```
data = pd.read_csv('amazon_prime_titles.csv')
```

In [3]:
```
data.head()
```

Out[3]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	The Grand Seduction	Don McKellar	Brendan Gleeson, Taylor Kitsch, Gordon Pinsent	Canada	March 30, 2021	2014	NaN	113 min	Comedy, Drama	A small fishing village must procure a local d...
1	s2	Movie	Take Care Good Night	Girish Joshi	Mahesh Manjrekar, Abhay Mahajan, Sachin Khedekar	India	March 30, 2021	2018	13+	110 min	Drama, International	A Metro Family decides to fight a Cyber Crimin...
2	s3	Movie	Secrets of Deception	Josh Webber	Tom Sizemore, Lorenzo Lamas, Robert LaSardo, R...	United States	March 30, 2021	2017	NaN	74 min	Action, Drama, Suspense	After a man discovers his wife is cheating on ...
3	s4	Movie	Pink: Staying True	Sonia Anderson	Interviews with: Pink, Adele, Beyoncé, Britney...	United States	March 30, 2021	2014	NaN	69 min	Documentary	Pink breaks the mold once again, bringing her ...
4	s5	Movie	Monster Maker	Giles Foster	Harry Dean Stanton, Kieran O'Brien, George Cos...	United Kingdom	March 30, 2021	1989	NaN	45 min	Drama, Fantasy	Teenage Matt Banting wants to work with a famo...

In [4]:

```
data.drop(['show_id'], axis = 1, inplace = True)
data.drop(['description'], axis = 1, inplace = True)
```

In [5]:

```
data.head()
```

Out[5]:

	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	Movie	The Grand Seduction	Don McKellar	Brendan Gleeson, Taylor Kitsch, Gordon Pinsent	Canada	March 30, 2021	2014	NaN	113 min	Comedy, Drama
1	Movie	Take Care Good Night	Girish Joshi	Mahesh Manjrekar, Abhay Mahajan, Sachin Khedekar	India	March 30, 2021	2018	13+	110 min	Drama, International
2	Movie	Secrets of Deception	Josh Webber	Tom Sizemore, Lorenzo Lamas, Robert LaSardo, R...	United States	March 30, 2021	2017	NaN	74 min	Action, Drama, Suspense
3	Movie	Pink: Staying True	Sonia Anderson	Interviews with: Pink, Adele, Beyoncé, Britney...	United States	March 30, 2021	2014	NaN	69 min	Documentary
4	Movie	Monster Maker	Giles Foster	Harry Dean Stanton, Kieran O'Brien, George Cos...	United Kingdom	March 30, 2021	1989	NaN	45 min	Drama, Fantasy

In [6]:

```
#Checking for duplicate
data.duplicated().sum()
```

Out[6]:

```
0
```

In [7]:

```
data.describe()
```

Out[7]:

	release_year
count	9668.000000
mean	2008.341849
std	18.922482
min	1920.000000
25%	2007.000000
50%	2016.000000
75%	2019.000000
max	2021.000000

In [8]:

```
data.info() # data.dtypes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9668 entries, 0 to 9667
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   type          9668 non-null   object
 1   title         9668 non-null   object
 2   director      7586 non-null   object
 3   cast          8435 non-null   object
 4   country       672 non-null    object
 5   date_added    155 non-null    object
 6   release_year  9668 non-null   int64 
 7   rating        9331 non-null   object
 8   duration      9668 non-null   object
 9   listed_in     9668 non-null   object
dtypes: int64(1), object(9)
memory usage: 755.4+ KB
```

In [9]:
```
data.isna().sum() #data.isnull().sum()
```

Out[9]:
```
type               0
title              0
director        2082
cast            1233
country         8996
date_added      9513
release_year       0
rating           337
duration           0
listed_in          0
dtype: int64
```

In [10]:
```
#Filling the null values
data['director'].fillna('Unavailable', inplace = True)
data['cast'].fillna('Unavailable', inplace = True)
data['country'].fillna('Unavailable', inplace = True)
data['date_added'] = data['date_added'].ffill()
data['rating'] = data['rating'].fillna(data['rating'].mode()[0])
```

In [11]:
```
data.isna().sum()
```

Out[11]:
```
type            0
title           0
director        0
cast            0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
dtype: int64
```

In [12]:
```
data.head()
```

Out[12]:

	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	Movie	The Grand Seduction	Don McKellar	Brendan Gleeson, Taylor Kitsch, Gordon Pinsent	Canada	March 30, 2021	2014	13+	113 min	Comedy, Drama
1	Movie	Take Care Good Night	Girish Joshi	Mahesh Manjrekar, Abhay Mahajan, Sachin Khedekar	India	March 30, 2021	2018	13+	110 min	Drama, International
2	Movie	Secrets of Deception	Josh Webber	Tom Sizemore, Lorenzo Lamas, Robert LaSardo, R...	United States	March 30, 2021	2017	13+	74 min	Action, Drama, Suspense
3	Movie	Pink: Staying True	Sonia Anderson	Interviews with: Pink, Adele, Beyoncé, Britney...	United States	March 30, 2021	2014	13+	69 min	Documentary
4	Movie	Monster Maker	Giles Foster	Harry Dean Stanton, Kieran O'Brien, George Cos...	United Kingdom	March 30, 2021	1989	13+	45 min	Drama, Fantasy

In [ ]:
```
data['date_added'].unique()
```

In [4]:
```
s1="March 30,2021"
s1.split()[1].split(",")
```

Out[4]:
```
['30', '2021']
```

In [13]:
```
data['date_added'] = pd.to_datetime(data['date_added'])
```

In [14]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9668 entries, 0 to 9667
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   type          9668 non-null   object
 1   title         9668 non-null   object
 2   director      9668 non-null   object
 3   cast          9668 non-null   object
 4   country       9668 non-null   object
 5   date_added    9668 non-null   datetime64[ns]
 6   release_year  9668 non-null   int64
 7   rating        9668 non-null   object
 8   duration      9668 non-null   object
 9   listed_in     9668 non-null   object
dtypes: datetime64[ns](1), int64(1), object(8)
memory usage: 755.4+ KB
```

In [15]:
```
data.head()
```

Out[15]:

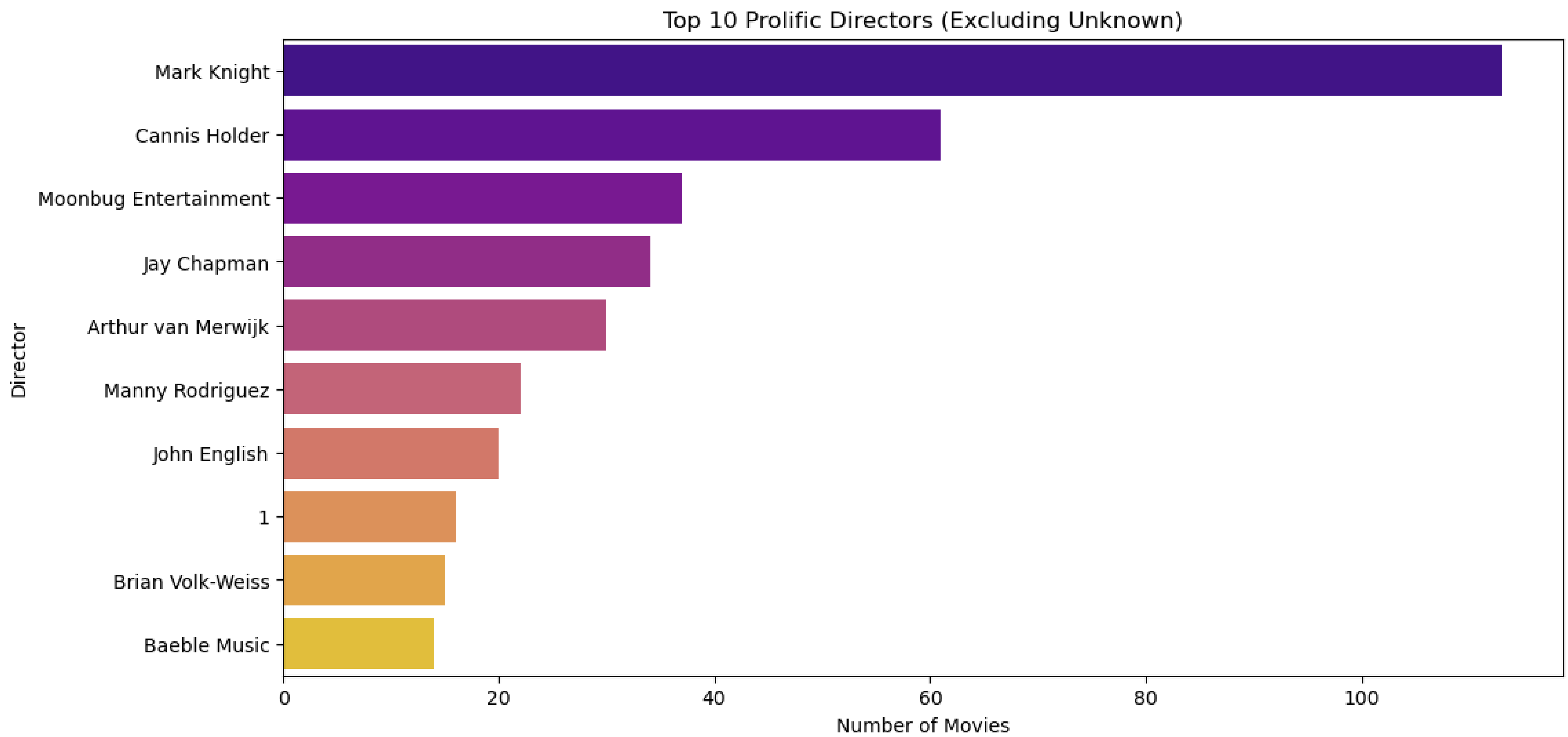
	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	Movie	The Grand Seduction	Don McKellar	Brendan Gleeson, Taylor Kitsch, Gordon Pinsent	Canada	2021-03-30	2014	13+	113 min	Comedy, Drama
1	Movie	Take Care Good Night	Girish Joshi	Mahesh Manjrekar, Abhay Mahajan, Sachin Khedekar	India	2021-03-30	2018	13+	110 min	Drama, International
2	Movie	Secrets of Deception	Josh Webber	Tom Sizemore, Lorenzo Lamas, Robert LaSardo, R...	United States	2021-03-30	2017	13+	74 min	Action, Drama, Suspense
3	Movie	Pink: Staying True	Sonia Anderson	Interviews with: Pink, Adele, Beyoncé, Britney...	United States	2021-03-30	2014	13+	69 min	Documentary
4	Movie	Monster Maker	Giles Foster	Harry Dean Stanton, Kieran O'Brien, George Cos...	United Kingdom	2021-03-30	1989	13+	45 min	Drama, Fantasy

In [16]:
```
#top 10 diretors with most directed movies
filtered_directors = data[data['director'] != 'Unavailable']
top_directors = filtered_directors['director'].value_counts()[:10]
```

In [ ]:
```
top_directors
```

In [17]:
```
plt.figure(figsize=(12, 6))
sns.barplot(x=top_directors.values, y=top_directors.index, palette='plasma')
```

```python
plt.title('Top 10 Prolific Directors (Excluding Unvailable)')
plt.xlabel('Number of Movies')
plt.ylabel('Director')
plt.show()
```



In [ ]:

```python
data["gender"].value_counts()

# Male  34
#Female 36
```

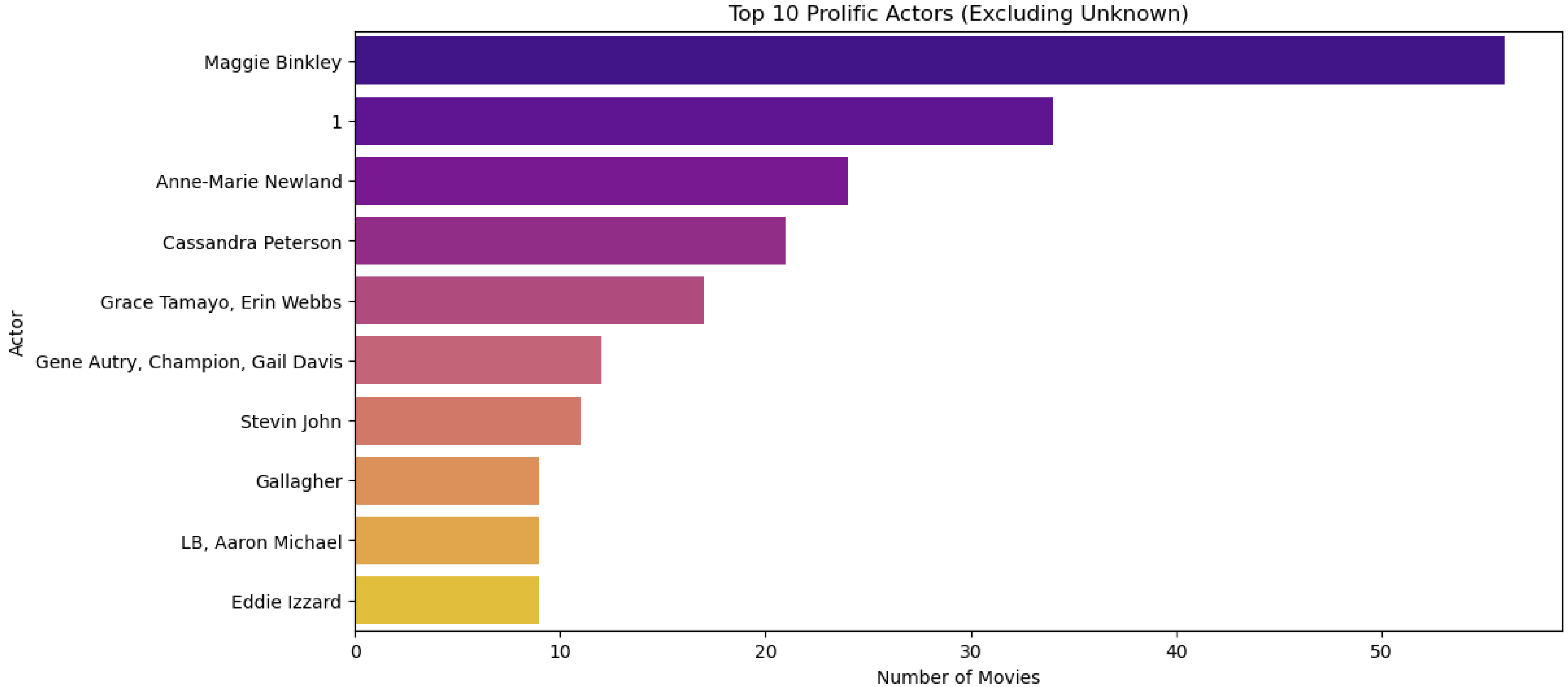
In [18]:

```python
filtered_actors = data[data['cast'] != 'Unavailable']
top_actors = filtered_actors['cast'].value_counts().head(10)
```

In [19]:

```python
plt.figure(figsize=(12, 6))
sns.barplot(x=top_actors.values, y=top_actors.index, palette='plasma')
plt.title('Top 10 Prolific Actors (Excluding Unknown)')
plt.xlabel('Number of Movies')
```

```
plt.ylabel('Actor')
plt.show()
```

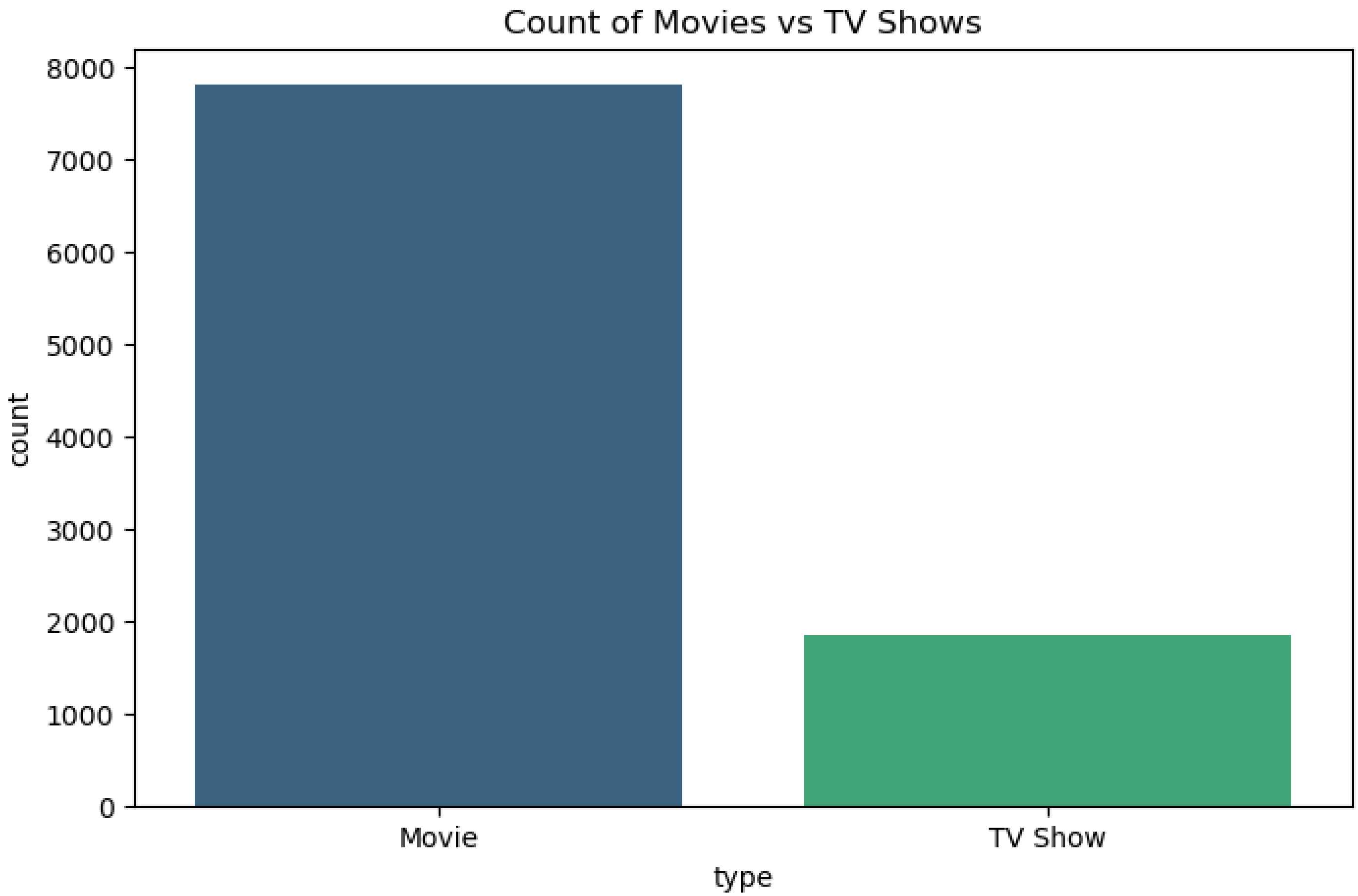


In [ ]:

```
data["type"].value_counts()
```

In [20]:

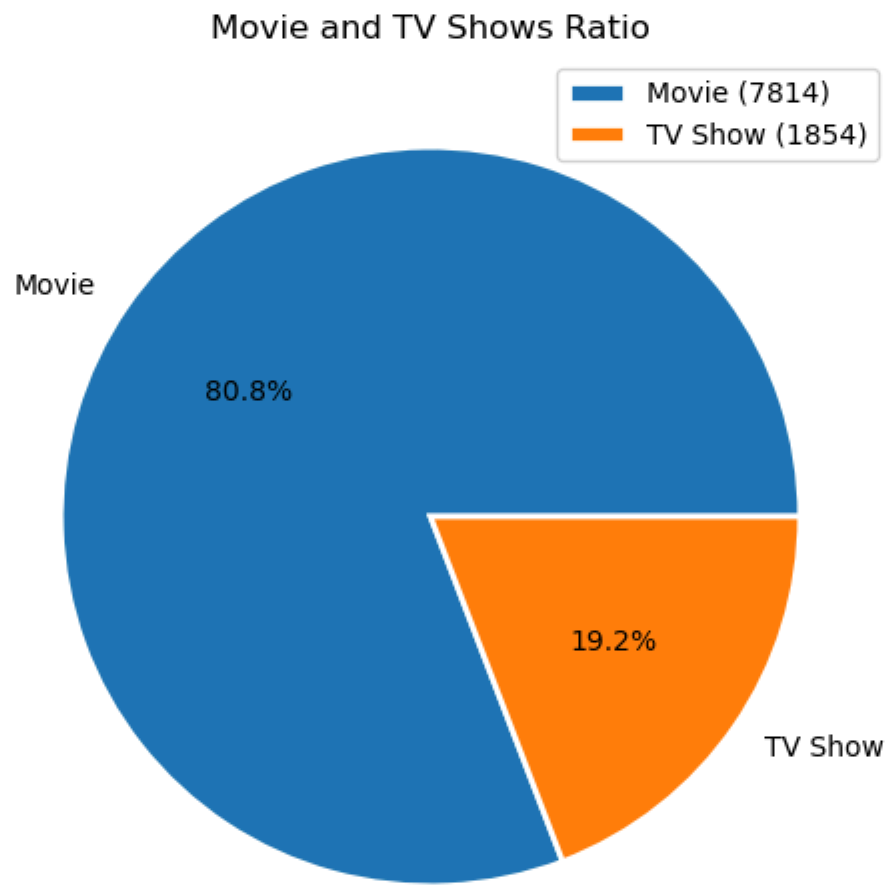
```
# Count of each content type (Movie vs. TV Show)
plt.figure(figsize=(8, 5))
sns.countplot(data=data, x='type', palette='viridis')
plt.title('Count of Movies vs TV Shows')
plt.show()
```



In [ ]:
```
data["type"].value_counts().plot(kind="pie")
```

In [ ]:
```
plt.pie(data["type"].value_counts(),labels=data["type"].value_counts().index,autopct="%1.2f%%"
       explode=[0.2,0])
plt.show()
```

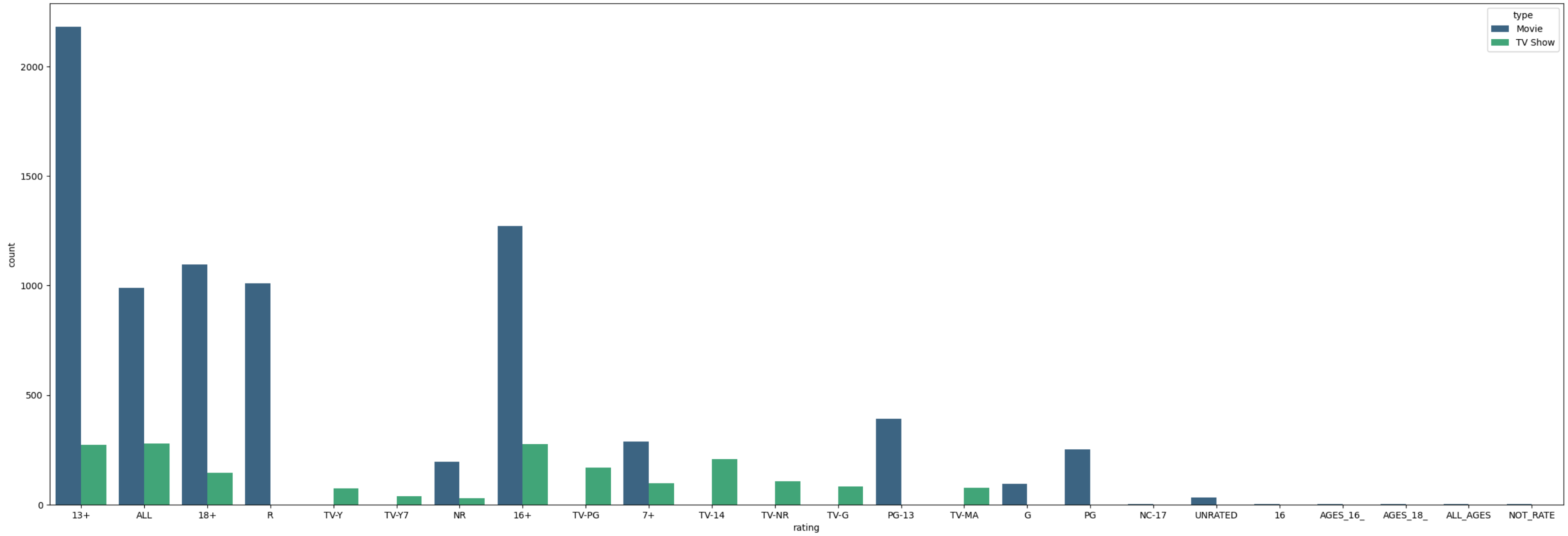
In [21]:
```
df_pie = data['type'].copy().value_counts()
plt.figure(figsize=(6,6))
plt.pie(x=df_pie.values,labels=df_pie.index,autopct='%1.1f%%' )
plt.title('Movie and TV Shows Ratio')
plt.legend()
plt.show()
```



In [ ]:
```
data['rating'].unique()
```

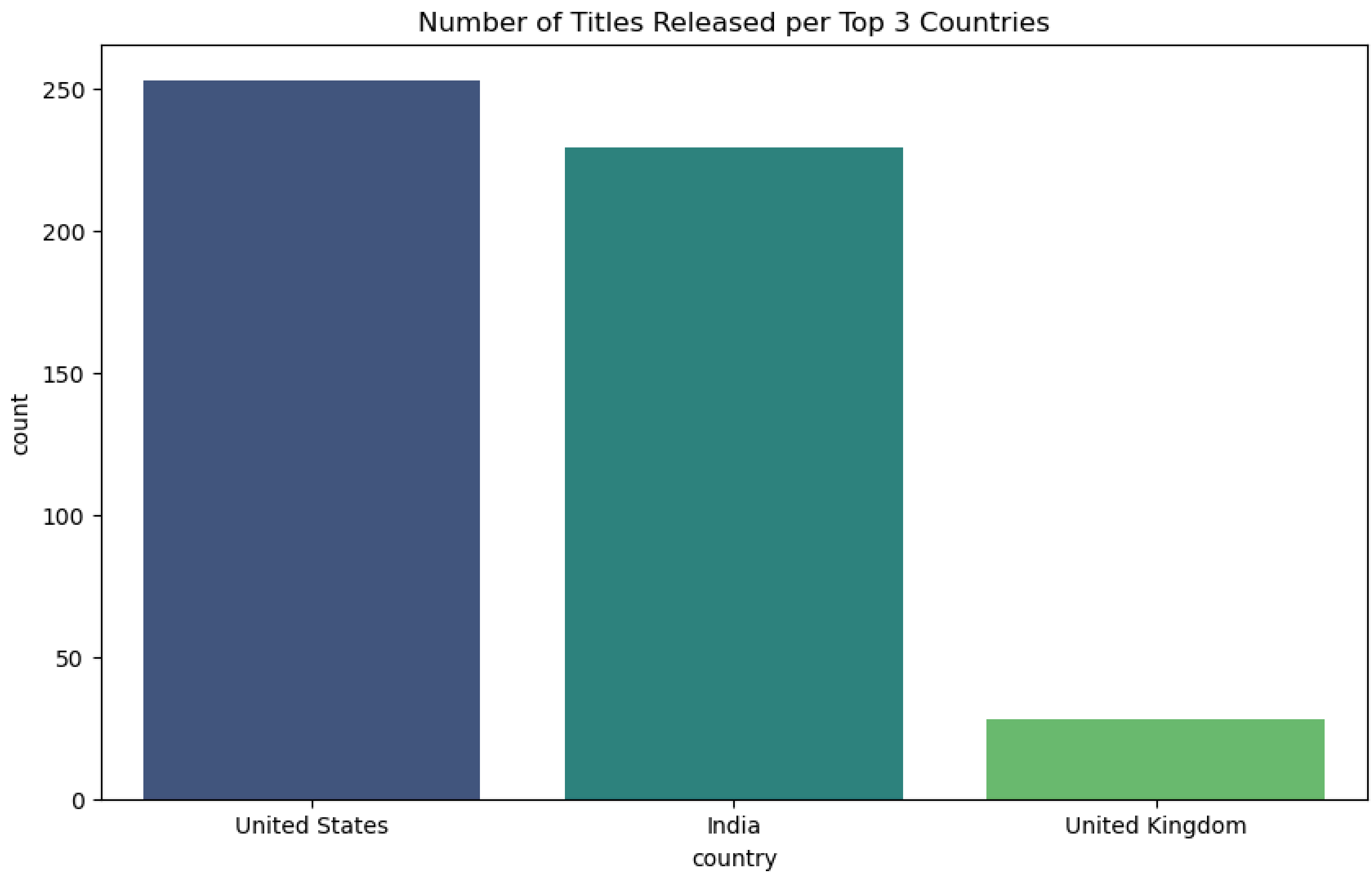
In [ ]:
```
data.groupby('type')["rating"].value_counts()
```

In [22]:
```
plt.figure(figsize=(30,10))
sns.countplot(x= data['rating'],data=data,hue = 'type',palette='viridis')
```



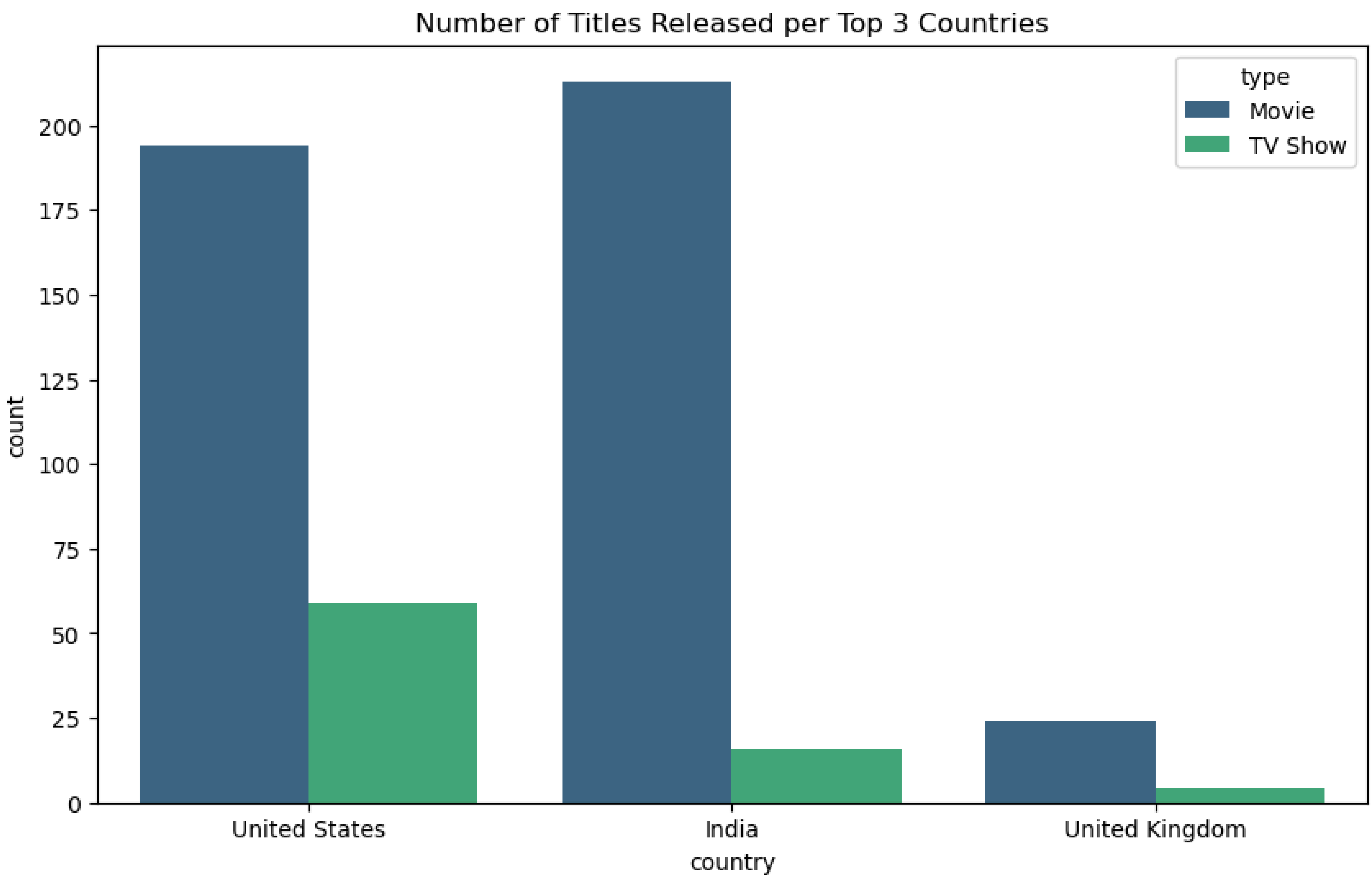
In [23]:

```
# Calculating the number of titles per country
filtered_country = data[data['country'] != 'Unavailable']
country_counts = filtered_country['country'].value_counts()

# Getting the top 3 countries
top_countries = country_counts.nlargest(3).index

# Filtering the DataFrame to include only the top 3 countries
df_top_countries = filtered_country[filtered_country['country'].isin(top_countries)]

# Plotting the number of titles released per top 3 countries
plt.figure(figsize=(10, 6))
sns.countplot(data=df_top_countries, x='country', palette='viridis')
plt.title('Number of Titles Released per Top 3 Countries')
plt.show()
```



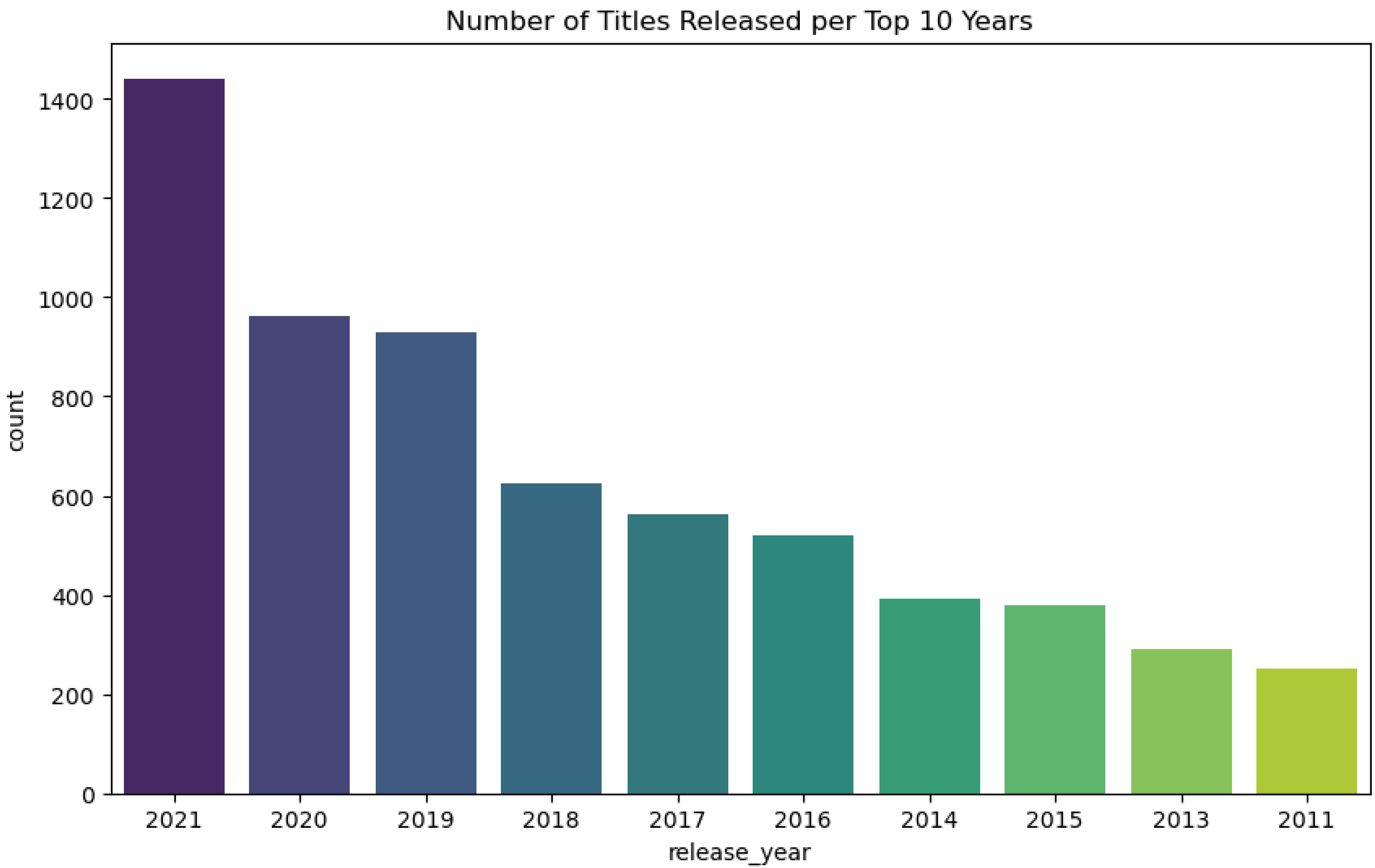
In [24]:
```
# Calculating the number of titles per country
filtered_country = data[data['country'] != 'Unvailable']
country_counts = filtered_country['country'].value_counts()

# Getting the top 3 countries
top_countries = country_counts.nlargest(3).index

# Filtering the DataFrame to include only the top 3 countries
df_top_countries = filtered_country[filtered_country['country'].isin(top_countries)]
```

```
# Plotting the number of titles released per top 3 countries
plt.figure(figsize=(10, 6))
sns.countplot(data=df_top_countries, x='country', hue = 'type', palette='viridis', order=top_countries)
plt.title('Number of Titles Released per Top 3 Countries')
plt.show()

#df_top_countries.groupby("type")["country"].value_counts()
```



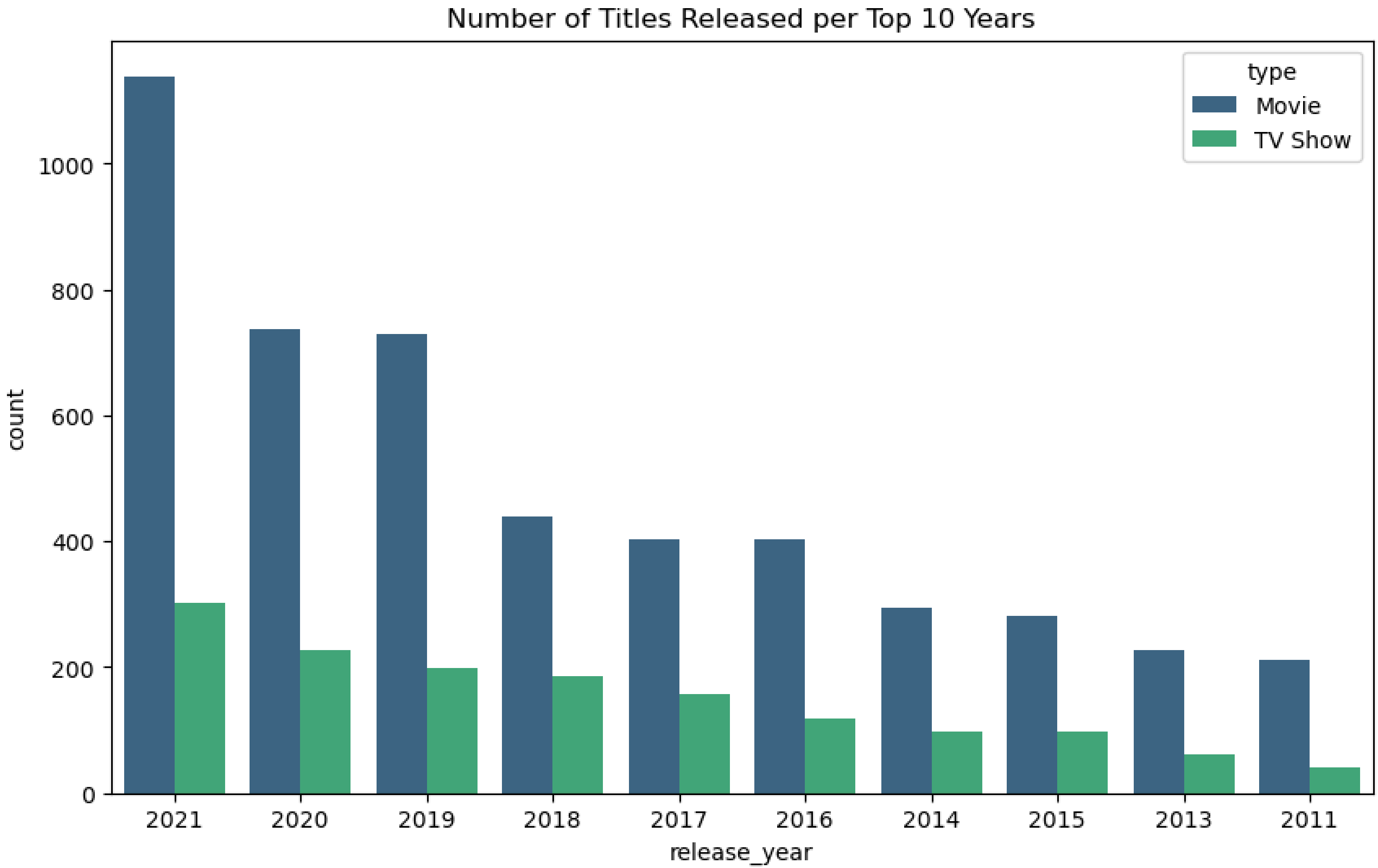
In [25]:
```
year_counts = data['release_year'].value_counts()

# Getting the top 10 release-year
```

```python
top_years = year_counts.nlargest(10).index

# Filtering the DataFrame to include only the top 10 Years
df_top_years = data[data['release_year'].isin(top_years)]

# Plotting the number of titles released per top 10 Years
plt.figure(figsize=(10, 6))
sns.countplot(data=df_top_years, x='release_year', palette='viridis', order=top_years)
plt.title('Number of Titles Released per Top 10 Years')
plt.show()
```



In [26]:

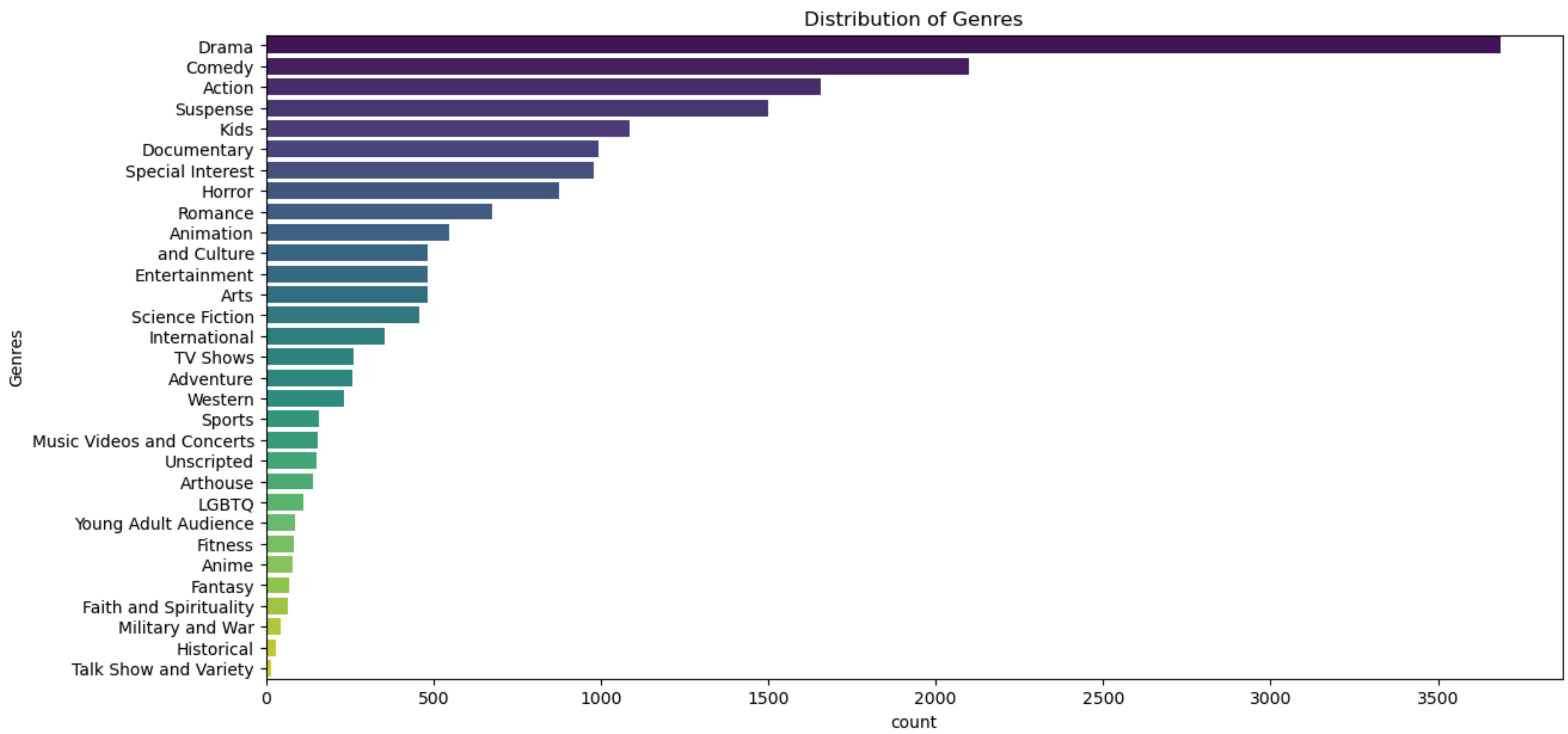
```
year_counts = data['release_year'].value_counts()

# Getting the top 10 release-year
top_years = year_counts.nlargest(10).index

# Filtering the DataFrame to include only the top 10 Years
df_top_years = data[data['release_year'].isin(top_years)]

# Plotting the number of titles released per top 10 Years
plt.figure(figsize=(10, 6))
sns.countplot(data=df_top_years, x='release_year', hue = 'type', palette='viridis', order=top_years)
plt.title('Number of Titles Released per Top 10 Years')
plt.show()
```



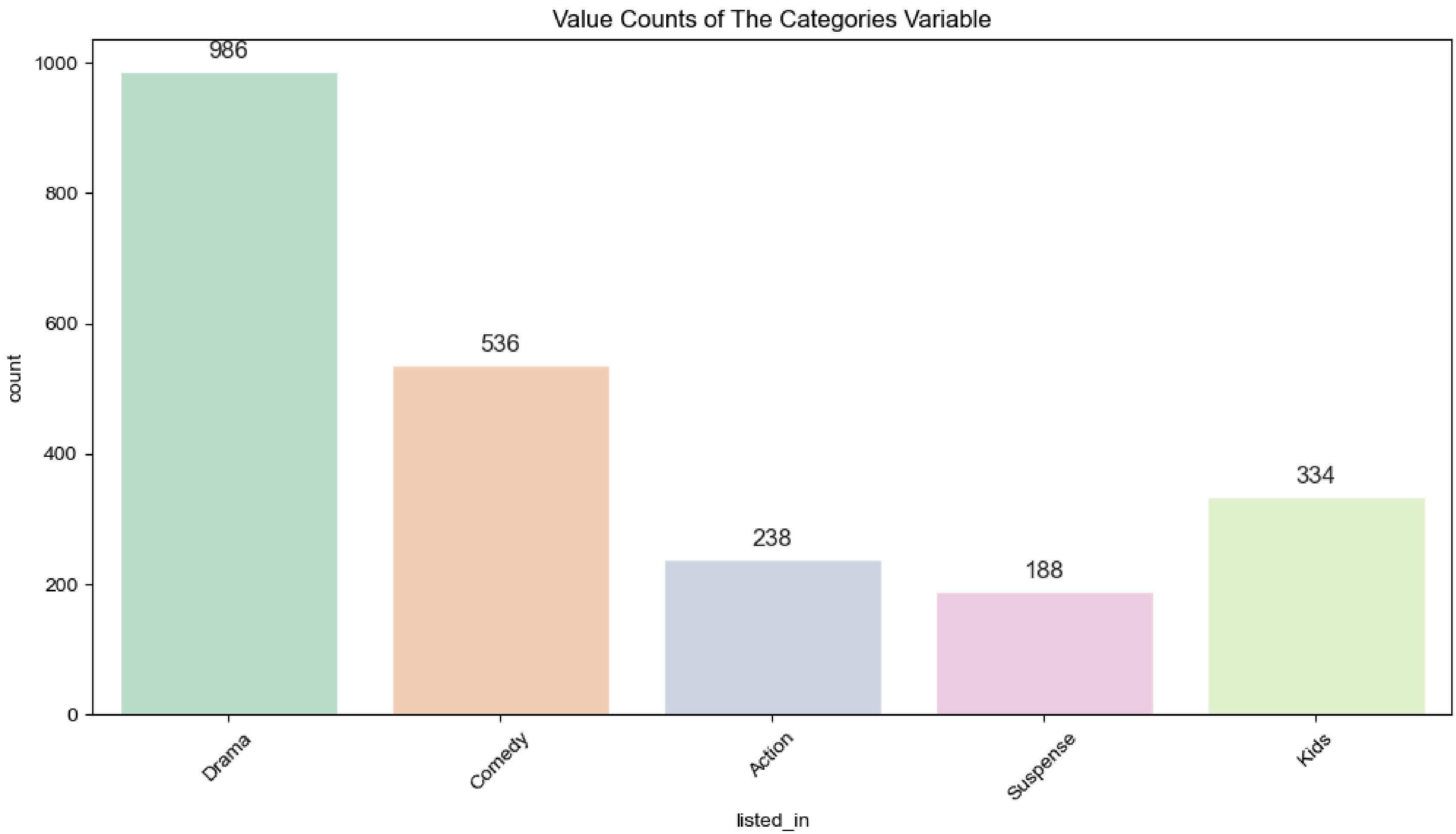
In [27]:

```
# Distribution of genres
# Splitting the genres
data['Genres'] = data['listed_in'].apply(lambda x: x.split(', '))

# Exploding the genres into separate rows
genres_exploded = data.explode('Genres')

plt.figure(figsize=(14, 7))
sns.countplot(data=genres_exploded, y='Genres',
              order=genres_exploded['Genres'].value_counts().index,
```

```
                  palette='viridis')
plt.title('Distribution of Genres')
plt.show()
```

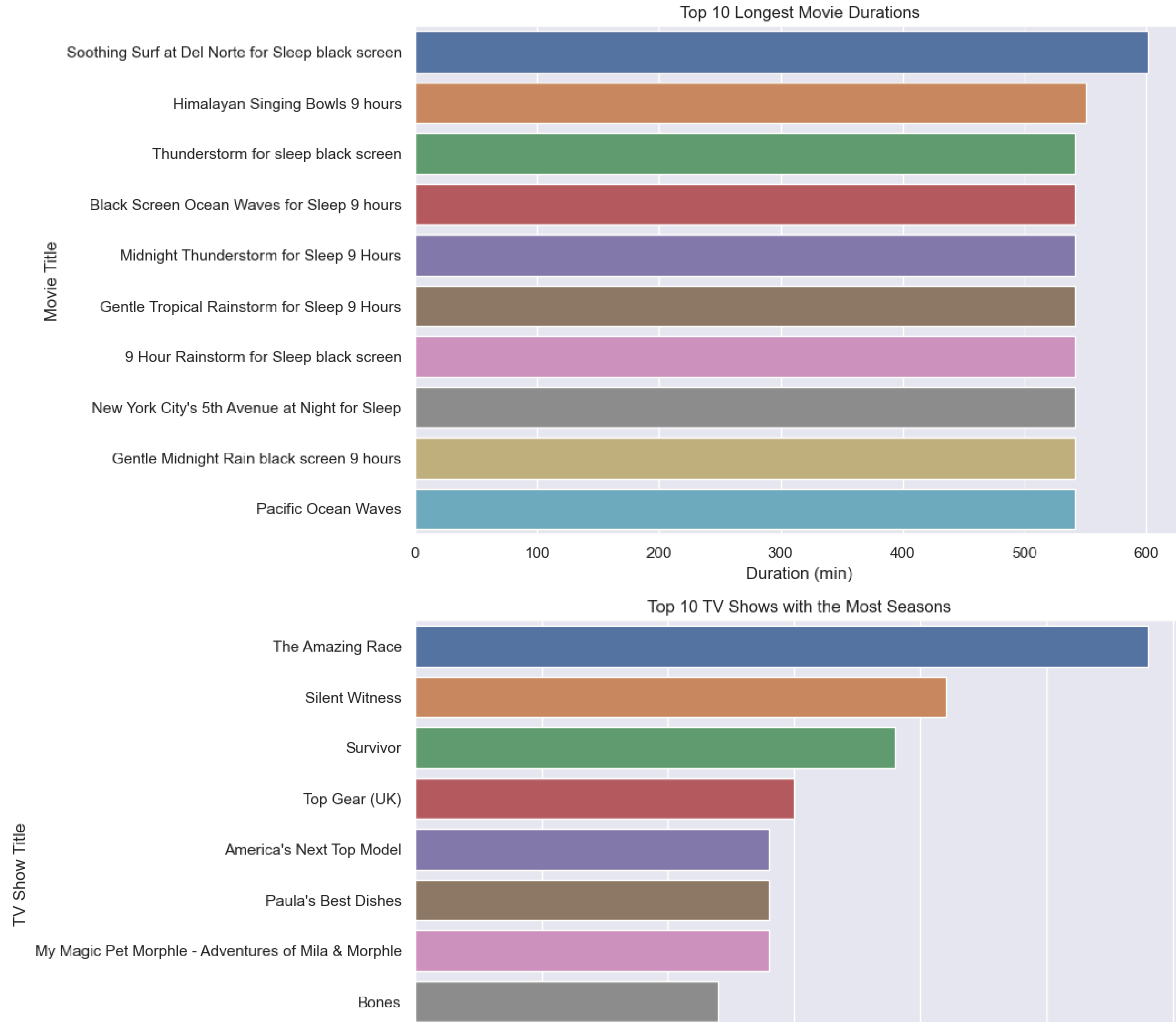


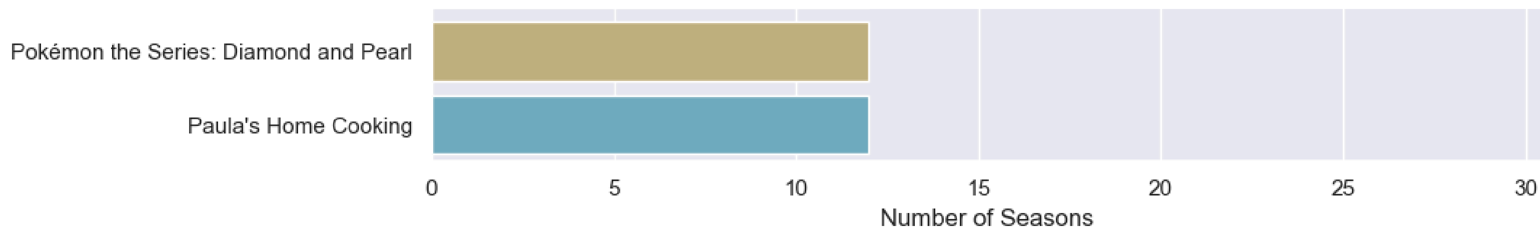
In [28]:
```
data['Genres'] = data['listed_in'].apply(lambda x: x.split(', '))
genres_exploded = data.explode('Genres')

plt.figure(figsize=(12, 6))
plt.title("Value Counts of The Categories Variable")
sns.set(style="darkgrid")
sns.countplot(x="listed_in", data=genres_exploded, palette="Pastel2",
                        order=genres_exploded['Genres'].value_counts().iloc[:5].index)

plt.xticks(rotation=45)  # Rotate x-axis labels for better readability
plt.show()
```



In [29]:
```
# Filter data for movies and TV shows separately
df_movies = data[data['type'] == 'Movie'].copy()
df_tv_shows = data[data['type'] == 'TV Show'].copy()

# Extract numeric duration for movies
df_movies['duration_num'] = df_movies['duration'].str.extract('(\d+)', expand=False).astype(float)

# Extract numeric duration for TV shows
df_tv_shows['duration_num'] = df_tv_shows['duration'].str.extract('(\d+)', expand=False).astype(float)
```

```python
# Plotting
plt.figure(figsize=(12, 12))

# Subplot 1: Top 10 Longest Movie Duration
plt.subplot(2, 1, 1)
sns.barplot(data=df_movies.sort_values(by='duration_num',
                                       ascending=False).head(10), y='title', x='duration_num',
            orient='horizontal')
plt.title('Top 10 Longest Movie Durations')
plt.xlabel('Duration (min)')
plt.ylabel('Movie Title')

# Subplot 2: Top 10 TV Shows with the Most Seasons
plt.subplot(2, 1, 2)
sns.barplot(data=df_tv_shows.sort_values(by='duration_num', ascending=False).head(10),
            y='title',
            x='duration_num', orient='horizontal')
plt.title('Top 10 TV Shows with the Most Seasons')
plt.xlabel('Number of Seasons')
plt.ylabel('TV Show Title')

plt.tight_layout()
plt.show()
```


Top 10 Longest Movie Durations
Movie Title
Soothing Surf at Del Norte for Sleep black screen
Himalayan Singing Bowls 9 hours
Thunderstorm for sleep black screen
Black Screen Ocean Waves for Sleep 9 hours
Midnight Thunderstorm for Sleep 9 Hours
Gentle Tropical Rainstorm for Sleep 9 Hours
9 Hour Rainstorm for Sleep black screen
New York City's 5th Avenue at Night for Sleep
Gentle Midnight Rain black screen 9 hours
Pacific Ocean Waves
0
100
200
300
400
500
600
Duration (min)
Top 10 TV Shows with the Most Seasons
TV Show Title
The Amazing Race
Silent Witness
Survivor
Top Gear (UK)
America's Next Top Model
Paula's Best Dishes
My Magic Pet Morphle - Adventures of Mila & Morphle
Bones



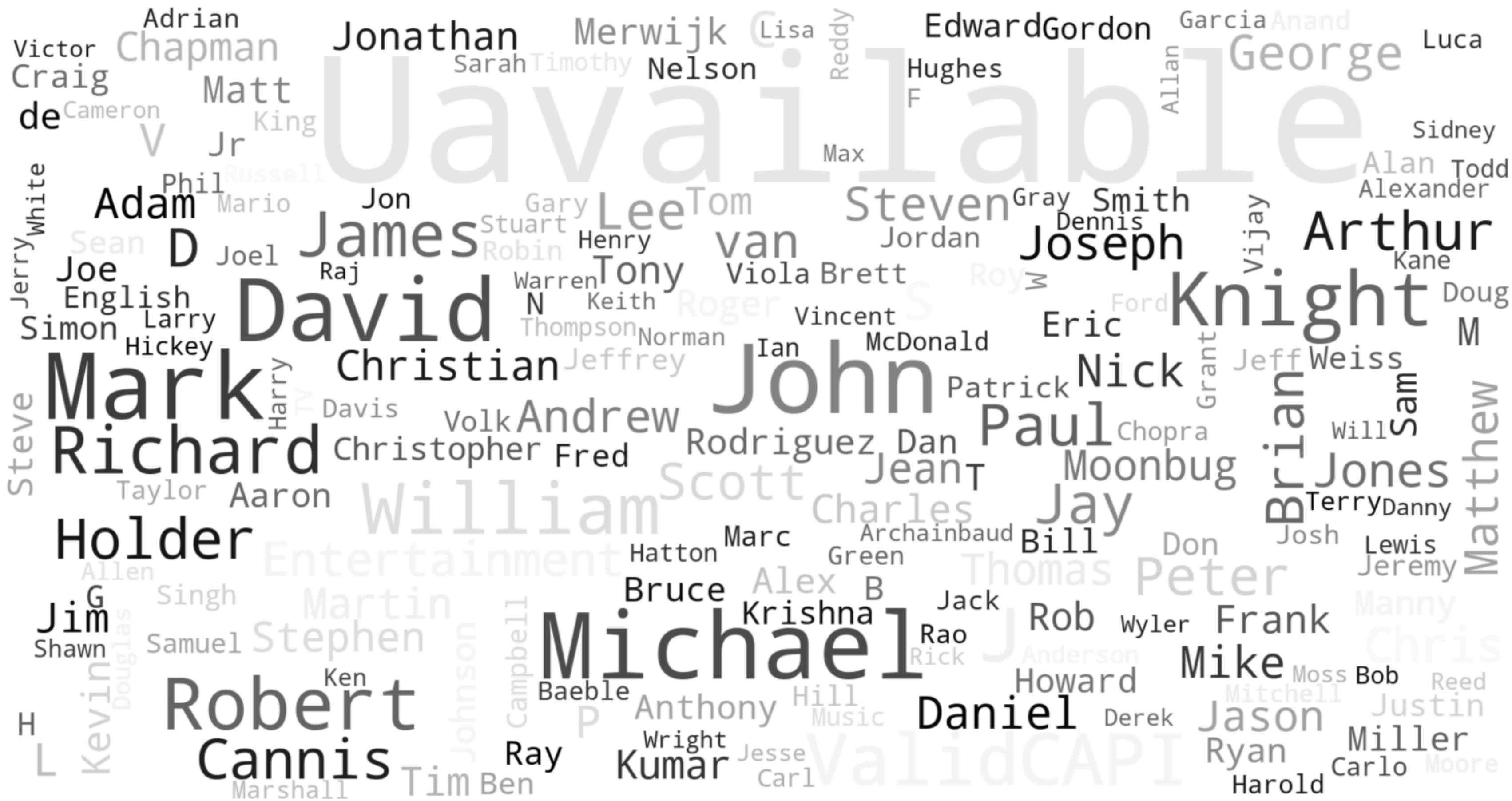

```
In [ ]: data["duration"].unique()
```

```
In [3]: s1="Shivaji23456"
        s1.extract('\d+')
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[3], line 2
      1 s1="Shivaji23456"
----> 2 s1.extract('\d+')

AttributeError: 'str' object has no attribute 'extract'
```

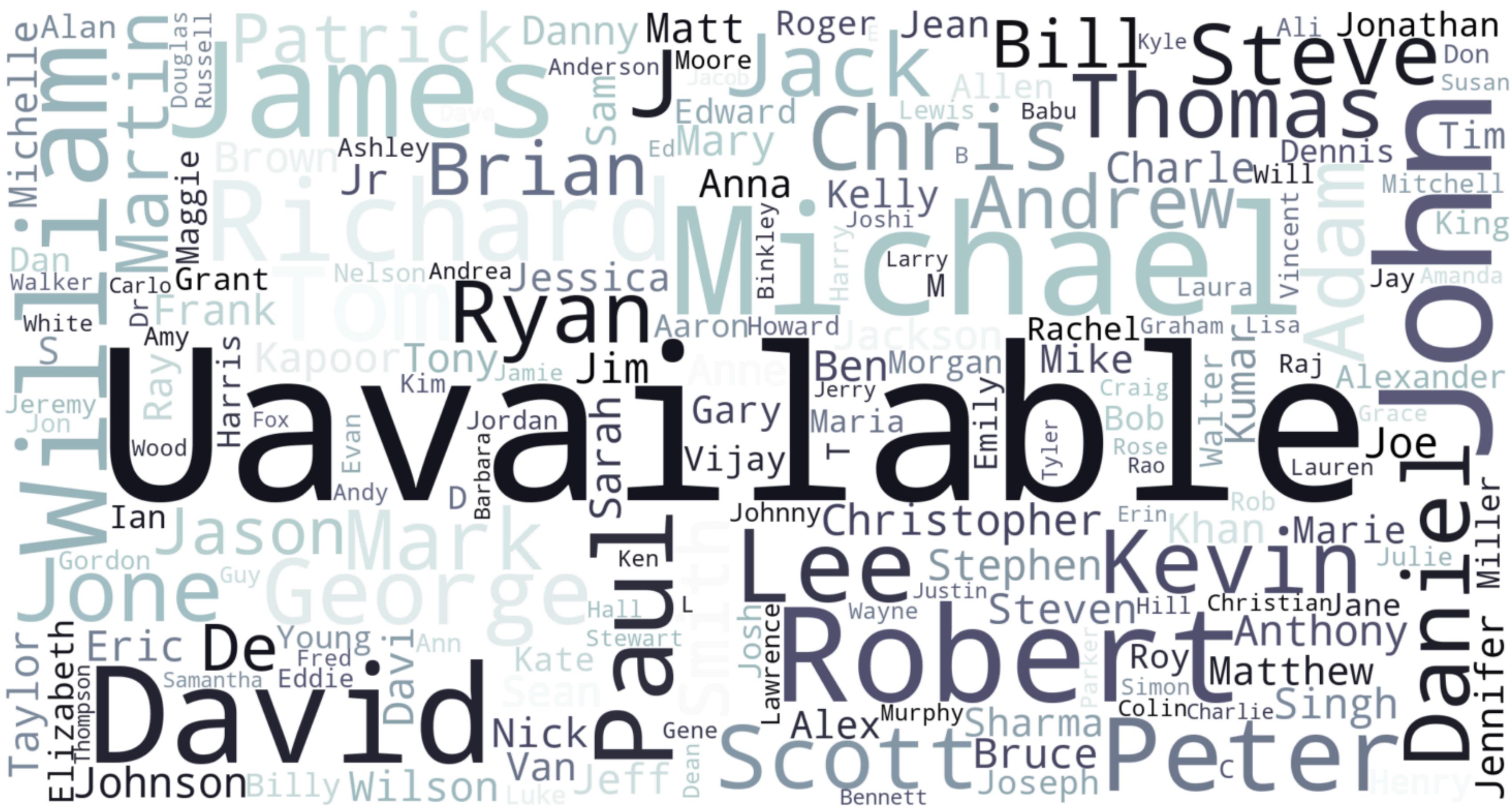
```
In [30]: txt1 = " ".join(title for title in data.title)

         word_cloud1 = WordCloud(collocations = False, background_color = 'white',
                                 width = 2048, height = 1080,colormap='Greens_r').generate(txt1)

         plt.figure(figsize=(30,10))
         plt.imshow(word_cloud1, interpolation='bilinear')
         plt.axis("off")
         plt.show()
```

In [31]:
```python
txt2 = " ".join(director for director in data.director)

word_cloud2 = WordCloud(collocations = False, background_color = 'white',
                        width = 2048, height = 1080,colormap='gist_gray').generate(txt2)

plt.figure(figsize=(30,10))
plt.imshow(word_cloud2, interpolation='bilinear')
plt.axis("off")
plt.show()
```



In [32]:

```
txt3 = " ".join(cast for cast in data.cast)

word_cloud3 = WordCloud(collocations = False, background_color = 'white',
                        width = 2048, height = 1080,colormap='bone_r').generate(txt3)

plt.figure(figsize=(30,10))
plt.imshow(word_cloud3, interpolation='bilinear')
plt.axis("off")
plt.show()
```


Uavailable
Michael
Robert
John
David
William
James
Richard
Peter
Scott
Lee
Paul
Steve
Thomas
Chris
Jack
Daniel
Kevin
Brian
Ryan
Mark
George
Jone
Jason
Tom
Andrew
Adam
Patrick
Martin
Bill
Matt
Roger
Jean
Jonathan
Alan
Danny
Michelle
Douglas
Russell
Anderson
Moore
Allen
Edward
Lewis
Babu
Mary
Ashley
Brown
Jr
Anna
Kelly
Charle
Will
Dennis
Tim
Mitchell
King
Don
Susan
Kyle
Ali
Maggie
Sam
Joshi
Binkley
Larry
Vincent
Laura
Jay
Amanda
Nelson
Andrea
Jessica
Grant
Carlo
Walker
Dan
Frank
White
Amy
Aaron
Howard
Jackson
Rachel
Graham
Lisa
Kapoor
Tony
Jim
Anne
Ben
Morgan
Mike
Raj
Alexander
Kim
Jamie
Harris
Jeremy
Jon
Fox
Jordan
Gary
Maria
Emily
Craig
Grace
Bob
Walter
Kumar
Wood
Evan
Barbara
Sarah
Vijay
Tyler
Rose
Rao
Joe
Lauren
Andy
Ian
Johnny
Christopher
Erin
Rob
Khan
Marie
Miller
Gordon
Guy
Stephen
Julie
Ken
Smith
Justin
Hall
Wayne
Steven
Hill
Christian
Jane
Eric
De
Young
Fred
Ann
Stewart
Josh
Anthony
Kate
Lawrence
Roy
Matthew
Taylor
Elizabeth
Thompson
Samantha
Eddie
Davi
Sean
Parker
Simon
Colin
Charlie
Singh
Nick
Gene
Alex
Murphy
Sharma
Jennifer
Johnson
Billy
Wilson
Van
Luke
Jeff
Dean
Bruce
Joseph
Bennett
Henry

In [ ]: